CSC1042 Computer Network II

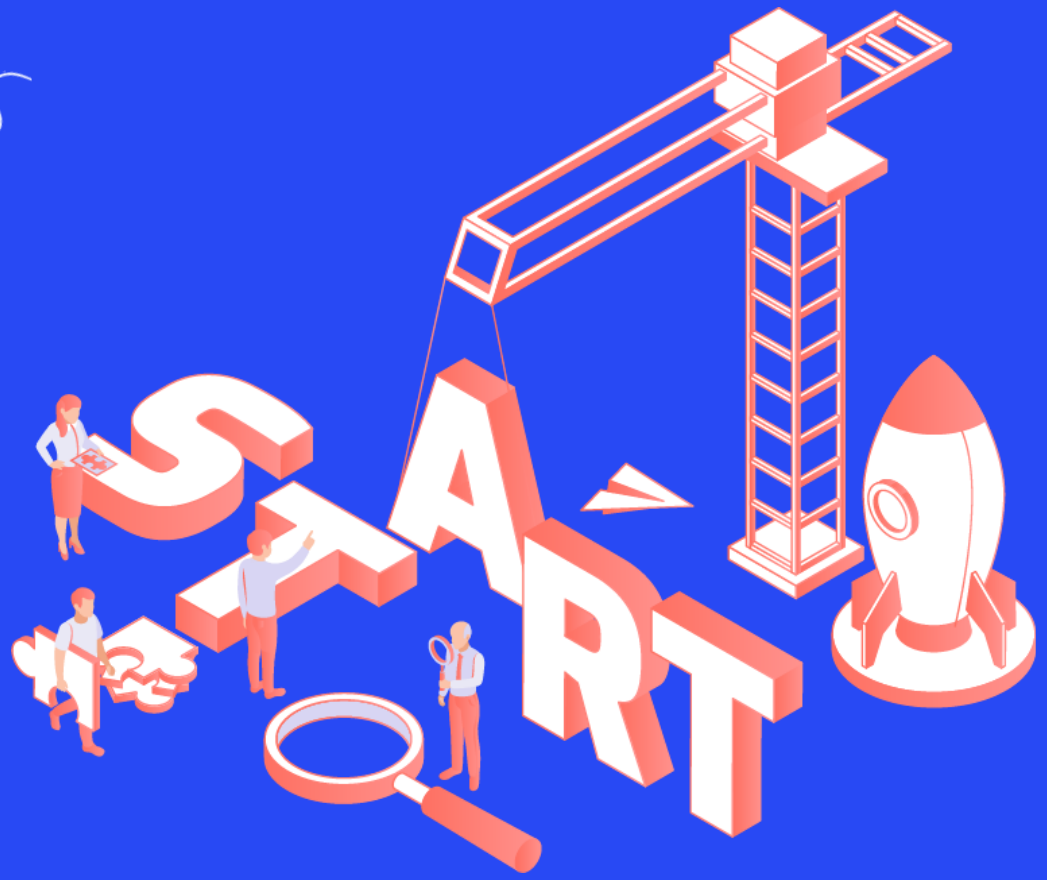
IPv4 Supernetting

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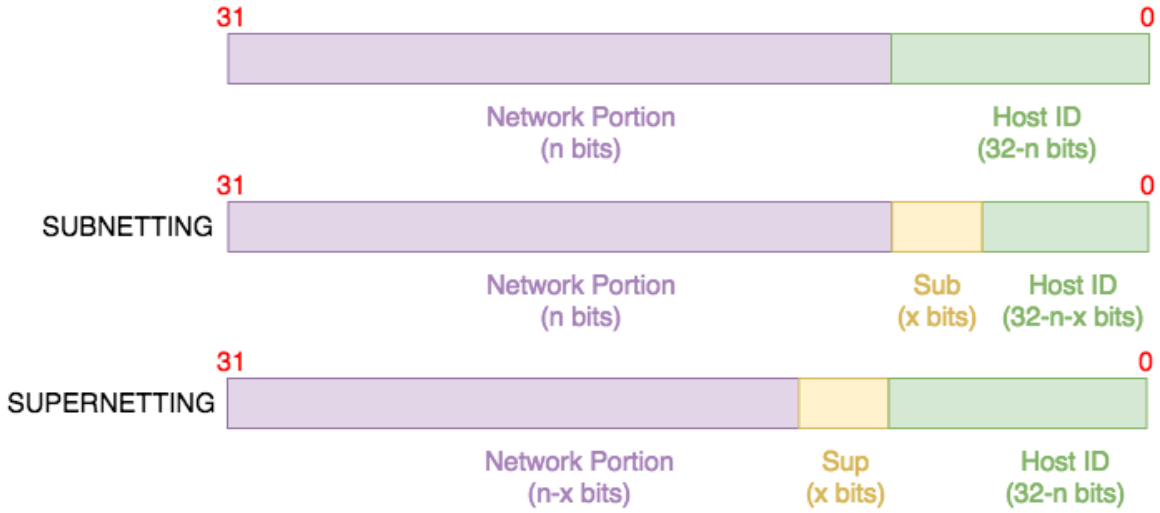
Learning outcomes

- Explain why supernetting is important
- Calculate IPv4 supernets



Supernetting

- Basic idea: creating a bigger network from small ones
- We do so by borrowing bits from the NetID
- The main purpose of supernetting is reducing the size of the routing tables



Rules to create supernets

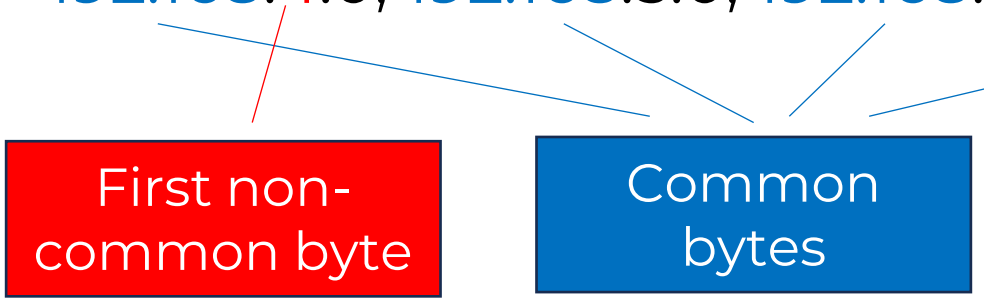
- Rule #1: Make sure the networks are contiguous

  192.168.0.0, 192.168.1.0, 192.168.2.0, 192.168.3.0 ✓

  192.168.10.0, 192.168.40.0, 192.168.55.0, 192.168.67.0 ✗

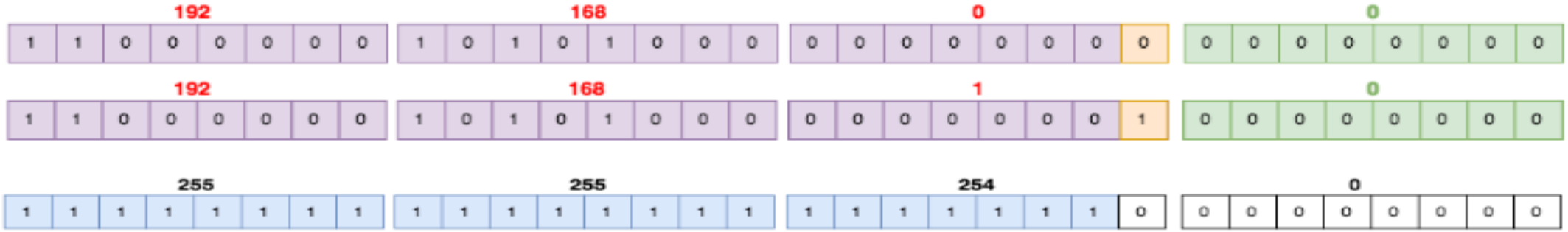
- Rule #2: the number of networks to be aggregated should be a power of 2 (i.e., 2, 4, 8, 16, …)
- Rule #3: The value of the first non-common byte in the first network IP address should be either 0 or a multiple of the number of networks to be aggregated

192.168.4.0, 192.168.5.0, 192.168.6.0, 192.168.7.0



How to supernet?

- Aggregate networks by:
  - Setting the common bits to 1 and setting the remaining bits to 0
- e.g., let's aggregate 192.168.0.0/24 and 192.168.1.0/24



- The supernet IP address will be the address of the 1st network in the list (192.168.0.0) with the new subnet mask 255.255.254.0 or 192.168.0.0/23

Supernet - example

- A company was assigned Class C networks having addresses from 192.4.16.0 up to 192.4.31.0. What is the supernet that can be created from these networks?

192.4.16.0 → 11000000.00000100.0001**0000**.00000000

192.4.31.0 → 11000000.00000100.0001**1111**.00000000

11111111.11111111.1111**0000**.00000000

- Subnet mask: 255.255.240.0 or /20 and IP address of the supernet is: 192.4.16.0/20
- Notice that 16 networks = 2^4, indicating that we borrowed 4 bits (in red) from the NetID, hence the /20

Supernetting - exercise

- Suppose an organization is allocated 4 contiguous Class C networks: 205.100.0.0, 205.100.1.0, 205.100.2.0, 205.100.3.0
  - What is the subnet mask of the supernet to be created?
  - What is the address of the supernet?

Supernetting - solution

205.100.0.0 → 11001101 . 01100100 . 00000000 . 00000000
205.100.1.0 → 11001101 . 01100100 . 00000001 . 00000000
205.100.2.0 → 11001101 . 01100100 . 00000010 . 00000000
205.100.3.0 → 11001101 . 01100100 . 00000011 . 00000000

11111111 . 11111111 . 111111**00**. 00000000

- Subnet mask: 255.255.252.0 or /22, IP address: 20.100.0.0/22
- Again, 4 networks = 2^2 → we borrowed 2 bits (red) from the NetID, hence /22

Supernetting - exercise

- Can both addresses appear in the routing table? Explain why.
  - 205.100.0.0/22
  - 205.100.0.0/20

Supernetting - solution

- 205.100.0.0 is a class C address → default mask is 255.255.255.0 or /24
- 205.100.0.0/22 → means that we borrowed 2 bits from NetID, so it contains $2^2 = 4$ class C networks

  205.100.0.0/22 → 11001101.01100100.0000 0000. 0000 0000

  205.100.1.0/22 → 11001101.01100100.0000 0001. 0000 0000

  205.100.2.0/22 → 11001101.01100100.0000 0010. 0000 0000

  205.100.3.0/22 → 11001101.01100100.0000 0011. 0000 0000

Supernetting - solution

- 205.100.0.0/20 → means that we borrowed 4 bits from NetID, so it contains 2^4 = 16 class C networks

  205.100.0.0/20 → 11001101.01100100.0000 0000. 0000 0000

  205.100.1.0/20 → 11001101.01100100.0000 0001. 0000 0000

  205.100.2.0/20 → 11001101.01100100.0000 0010. 0000 0000

  …

  205.100.15.0/20 → 11001101.01100100.0000 1111. 0000 0000

- Hence, all class C networks in 205.100.0.0/22 are included in 205.100.0.0/20
- Therefore, 205.100.0.0/20 is the only address to put in the routing table

Supernetting - exercise

- Given the address 192.20.4.0/22, answer the following questions:
  - What is the subnet mask in decimal?
  - What is the broadcast address?
  - How many class C networks in this supernet?
  - What are their addresses?
  - What are the addresses of the first and last host in this supernet?

Supernetting - solution

- Since it's /22, we know that there are 22 bit in NetID which should be set to 1 to get the subnet mask

  11111111.11111111.111111**00**.0000 0000 → 255.255.252.0

- Broadcast address
  - 192.20.4.0 → 11000000.00011000.000001**00**.**00000000**
  - Set bits in red to 1 to get the broadcast address

    11000000.00011000.000001**11**.**11111111** → 192.20.7.255
- /22 means that we borrowed 2 bits from NetID

  $2^2 = 4$ Class C networks

Supernetting - solution

- Their addresses:
  - 11000100.00010100.0000 0100. 0000 0000 (192.20.4.0/22)
  - 11000100.00010100.0000 0101. 0000 0000 (192.20.5.0/22)
  - 11000100.00010100.0000 0110. 0000 0000 (192.20.6.0/22)
  - 11000100.00010100.0000 0111. 0000 0000 (192.20.7.0/22)

- 1st and last host addresses of the supernet
  - 1st host: 192.20.4.1
  - Last host : 192.20.7.254

THANK YOU